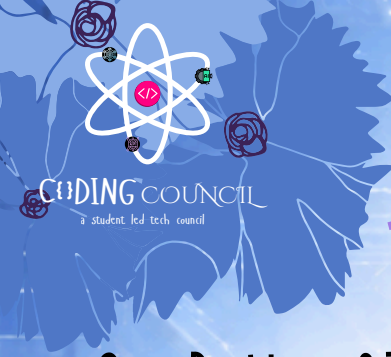




PYTHON WINTER SPRINT 2026



Core Problem of 7th Jan 2026 : Clinical Data Auditor

Each Day have one constant theme and three different tracks : Foundation Track especially due to W3B Collaboration, Core Track for those following since DAY-1 and the Project Track for those who have their basics of Python Cleared, they will make these end-to-end projects

FOUNDATION TRACK (FOR ABSOLUTE BEGINNERS AND NEW JOINERS)

[Expected time to Finish – 20 minutes]

Resource: <https://www.youtube.com/watch?v=UrsmFxEIp5k> from start till 1hr 40 minutes

Core Topic: Variables, Data types, Strings

Problem Statement

Write a Python program that:

1. Takes the following input from the user:
 - o Patient Name
 - o Age
 - o Heart Rate
 - o City
2. Perform simple string operations:
 - o Convert patient name to Title Case
 - o Convert city name to Uppercase
3. Print a clean Clinical Data Record in a formatted way.

⚠ This task is only about recording data, not validating or auditing it.

Submission Guidelines

Submit Before 11 pm 7th January

GitHub repository is mandatory - Commit history should reflect real work

Optionally, Include a short README explaining: What the system does, Audit logic and Disclaimer

Sample Input:

```
Enter patient name: ali khan
Enter age: 21
Enter heart rate: 78
Enter city: delhi
```

Sample Output:

```
-----
CLINICAL DATA RECORD
-----
Patient Name : Ali Khan
Age          : 21
Heart Rate   : 78 bpm
City         : DELHI
-----
```

Note: This record is for data entry only.

Beginners Please ask in Group if you get to encounter any problem, any kind of problem in this, or just DM me
Every One was once a beginner !!



PYTHON WINTER SPRINT 2026



CORE TRACK (FOR STUDENTS WHO HAVE BEEN LEARNING FROM DAY 1)

Core Topic: Functions and Recursion

Resource: <https://www.youtube.com/watch?v=UrsmfxEIp5k> from 4hr 50 minutes till 5hr 54 minutes

[Expected Time to Finish - 30 minutes]

Problem Statement

Write a Python program that audits basic clinical data using functions and recursion.

The program should:

Take patient details:

- Name
- Age
- Heart rate
- Number of vitals readings

Use functions to:

- Validate age
- Validate heart rate
- Decide audit status

Use recursion to:

- Take multiple heart rate readings (one by one)

Audit Logic:

- Age must be between 0-120
- Heart rate must be numeric
- If any heart rate > 180 → Warning
- If invalid input → Flag

Final Audit Status:

- FAIL → Any flag
- REVIEW → No flags, but warnings
- PASS → Clean data

Submission Guidelines

Submit Before 11 pm 7th January

GitHub repository is mandatory - Commit history should reflect real work

Optionally, Include a short README explaining: What the system does, Audit logic and Disclaimer

Sample Input:

```
Enter patient name: Ayaan Khan
Enter age: 22
Enter number of heart rate readings: 3

Enter heart rate reading: 76
Enter heart rate reading: 185
Enter heart rate reading: 82
```

Sample Output:

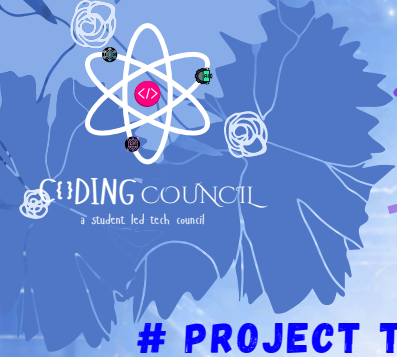
```
--- CLINICAL AUDIT RESULT ---
Patient : Ayaan Khan
Status  : REVIEW
Flags   : None
Warnings: ['High heart rate detected']
```

Sample Input (With Flag Case)

```
Enter patient name: Sara Ali
Enter age: twenty
Enter number of heart rate readings: 2

Enter heart rate reading: 72
Enter heart rate reading: 80
```

```
--- CLINICAL AUDIT RESULT ---
Patient : Sara Ali
Status  : FAIL
Flags   : ['Invalid age']
Warnings: None
```

PYTHON WINTER SPRINT 2026



PROJECT TRACK (FOR ADVANCED PARTICIPANTS, MINI PROJECT)

[Expected Time to Finish – 3-4 hours]

🏥 Clinical Data Audit System

Project Instructions (Advanced Track)

🎯 Objective

Build a CLI-based Clinical Data Audit System that audits patient-provided clinical data for safety, validity, and consistency. The system must be non-diagnostic and focus only on data auditing, similar to backend validation layers used in real healthcare software.

📥 User Inputs

Your program must collect the following:

- Patient Name
- Age
- Heart Rate (bpm)
- Blood Pressure:
 - Systolic
 - Diastolic
- Allergy information (yes / no)

🔍 Audit Rules (Mandatory)

VALIDATION RULES

- Age must be a number and within 0-120
- Heart rate must be numeric
- Blood pressure values must be numeric
- Systolic BP must be greater than diastolic BP
- Allergy field must be either yes or no

SAFETY RULES (WARNINGS)

- Heart rate outside 40-180 bpm → Warning
- Systolic BP outside 70-200 → Warning
- Diastolic BP outside 40-130 → Warning

📄 Audit Classification

Based on audit results, assign one status:

- PASS → No flags or warnings
- REVIEW → Warnings present but no hard failures
- FAIL → Any validation rule violated

🔑 Traceability Requirements

- Generate a random unique Audit ID for every audit for further traceability
- Attach the Audit ID to:
 - Console output
 - File logs

📌 Submission Guidelines

Submit Before 11 pm 7th January

GitHub repository is mandatory - Commit history should reflect real work

Optionally, Include a short README explaining: What the system does, Audit logic and Disclaimer

PROJECT TRACK (FOR ADVANCED, SYSTEM-LEVEL PROBLEM SOLVING)

File I/O Requirements

Create an append-only audit log file, Each audit record must include: Audit ID, Timestamp, Patient name, Audit status (Pass/Review/Fail), Flags (if any validation rule fails), Warnings (if any)

- Older audit records must never be overwritten

Output Requirements

Display a clean Audit Report in the console containing:

1. Audit ID
2. Timestamp
3. Patient name
4. Audit status (Pass/Review/Fail)
5. Flags (Failure of Validation Rules)
6. Warnings
7. Disclaimer that this is a non diagnostic Report Only, Not medically certified

Strict Constraints

-  No diagnosis
-  No treatment recommendations
-  No medical advice
-  Audit & validation only

Sample Run:

```
Program Start

yaml

=== Clinical Data Audit System ===
Enter patient name: arshad ali
Enter age: 22
Enter heart rate (bpm): 185
Enter systolic BP: 118
Enter diastolic BP: 76
Any known allergies? (yes/no): no

Console Output

yaml

=====
CLINICAL AUDIT REPORT
=====
Audit ID       : AUD-483921
Timestamp      : 2026-01-07 18:42:11
Patient Name   : Arshad Ali
Audit Status   : REVIEW

Audit Flags:
None

Warnings:
Heart rate outside typical safe range

Audit record saved for traceability.
This system performs data audit only.
No diagnosis or medical advice provided.
=====
```

Audit Log File (clinical_audit_log.txt)

```
yaml

=====
Audit ID       : AUD-483921
Timestamp      : 2026-01-07 18:42:11
Patient Name   : Arshad Ali
Audit Status   : REVIEW
Audit Flags: None
Warnings:
- Heart rate outside typical safe range
Note: Non-diagnostic clinical data audit
=====
```

Sample Run – FAIL Case

Input

```
yaml

Enter patient name: sara khan
Enter age: two
Enter heart rate (bpm): 78
Enter systolic BP: 110
Enter diastolic BP: 80
Any known allergies? (yes/no): yes
```

Output

```
yaml

Audit Status   : FAIL

Audit Flags:
Invalid age format

Warnings:
None
```